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EXPERIMENT NO. : 9 (ASSIGNMENT 4)

DATE: 16/10/2026

1. AIM & LEARNING OUTCOMES

Aim:

To study and compare the Metal-First and Metal-Last (Replacement Metal Gate / RMG) integration schemes in CMOS fabrication, analyze the impact of thermal budget constraints on gate metal diffusion using Semulator3D process modeling, and investigate high-k / metal gate (HKMG) integration in an advanced 14 nm FinFET logic process.

Learning Outcomes:

1. Understand the fundamental architectural differences between metal-first and metal-last (replacement gate) processes.

2. Comprehend the necessity of replacement metal gates due to thermal budget constraints during source/drain dopant activation anneals.

3. Analyze the critical role of conformal deposition and pinch-off challenges in high-aspect-ratio recessed gate trenches.

Starting Resources:

• Simulation Projects: Assignment4_Gatefirst.zam, Assignment4_GateLast.zam, and 14nmFinFET.zam.

• Literature References: Integrating High-K (Semiconductor Digest); High-K Materials and Metal Gates for CMOS Applications (J. Robertson & R.M. Wallace).

2. THEORETICAL BACKGROUND

1. Need for High-k / Metal Gate (HKMG):

In traditional sub-micron CMOS, polysilicon gates with silicon dioxide (SiO_2) dielectrics were standard. As devices scaled below 45 nm, gate oxide thinning caused excessive quantum mechanical tunneling leakage, while polysilicon suffered from poly-depletion effects that increased effective oxide thickness (EOT). HKMG stacks solved these issues by replacing SiO_2 with high-dielectric-constant materials (e.g., HfO_2) and polysilicon with work-function tuned metal gates.

2. Metal-First (Gate-First) Scheme:

In the Gate-First approach, the high-k dielectric and metal gate stack are deposited and patterned before the high-temperature (> 1000°C) source/drain dopant activation anneals. Consequently, the metal gate undergoes the full thermal budget, which causes metal atom diffusion into the dielectric and active channel. This causes threshold voltage (V_{th}) shifts, Fermi-level pinning, and device failure when contamination exceeds 10^{12} atoms/cm^3.

3. Metal-Last (Replacement Metal Gate / RMG) Scheme:

In the Gate-Last approach, a sacrificial (dummy) polysilicon gate is fabricated first to define the channel and self-align the source/drain junctions. After source/drain implantation and high-temperature anneals, an interlayer dielectric (ILD) is deposited and planarized via CMP. The dummy polysilicon gate is then selectively removed ('poly pull etch'), and the real metal gate stack is deposited into the resulting trench at low temperatures (< 450°C), preserving gate integrity.

3. QUESTION 1: GATE FIRST PROCESS ANALYSIS (ASSIGNMENT4_GATEFIRST.ZAM)

The Assignment4_Gatefirst.zam project was built and analyzed in Semulator3D. In this process flow, the gate stack is deposited before the high-temperature anneal steps for source and drain formation.

(a) Aluminum Metal Diffusion Plot & Active Region Concentration:

The high-temperature activation anneal causes metal diffusion across the gate oxide into the underlying silicon active region. Under default process conditions (12 s anneal), the simulated aluminum concentration in the active silicon channel reaches 1.24×10^{15} atoms/cm^3, far exceeding the operational limit of 10^{12} atoms/cm^3.

Table 1 outlines the complete Process Editor sequence for the Gate-First integration scheme.

Table 1: Process Editor Recipe Steps for Gate-First Flow (Assignment4_Gatefirst.zam)

Step	Process Stage	Step Name	Material / Chemistry	Operation & Purpose
1	Wafer Setup	Wafer Setup	Silicon Substrate	Defines p-type silicon wafer boundary and orientation
-1	Isolation Pad	Pad Oxide & SiN Hardmask	SiO_2 / Si_3N_4	Deposits stress relief pad oxide and nitride polish mask
-2	STI Trench	STI Lithography & Trench Etch	Photoresist / RIE Etch	Patterns and etches deep trenches into silicon substrate
-2	STI Fill & CMP	Thermal Ox, CVD Fill, CMP	Thermal SiO_2 / HDP Oxide	Passivates trench sidewalls, fills with oxide, and planarizes
-1	Mask Strip	SiN & Pad Oxide Strip	Hot H_3PO_4 / Dilute HF	Exposes clean active silicon surface for gate stack
-1	Gate Dielectric	Gate Oxide & High-k Deposition	SiO_2 (Interfacial) / HfO_2	Deposits thin interfacial oxide and conformal high-k dielectric
-2	NMOS Gate Metal	NMOS Metal & Gate Pattern	TiAl (NMOS) / TiN / Resist	Deposits low work-function NMOS metal and patterns for PMOS
-0.7	Gate Stack Etch	Final Gate Metal & Etch	W / TiN / TiAl / HfO_2	Deposits bulk metal and subtractively etches full gate stack
-0.5	Spacer & NMOS S/D	Spacer Deposition & NMOS Implant	Si_3N_4 Spacer / Arsenic	Forms sidewall spacer and implants NMOS source/drain dopants
-0.3	PMOS S/D Implant	PMOS Pattern & Boron Implant	Photoresist / BF_2 (Boron)	Selectively patterns and implants PMOS source/drain regions
19.5	Activation Anneal	Dopant Activation Diffusion	High Temperature (RTA)	High-temperature anneal causing severe metal cross-diffusion
-1	BEOL Contacts	ILD Deposition, Contact Etch, CMP	CVD Oxide / TiN / CMP	Passivates transistor, etches contact vias, and completes CMP

3. QUESTION 1 (B): MAXIMUM ANNEAL TIME BEFORE EXCEEDING LIMIT

(b) Maximum Anneal Time Determination:

The device stops functioning properly if metal contamination in the active region exceeds 10^{12} atoms/cm^3. To evaluate thermal limits, the post-implant anneal time was reduced in the recipe editor from 12.0 s down to 1.0 s.

Table 2 shows the simulated aluminum concentration across anneal times, and Figure 3 shows the concentration profile plotted on a logarithmic scale against anneal duration.

Key Finding:

At t = 6.12 s, the concentration is 1.54×10^6 atoms/cm^3 (safe). At t = 6.15 s, the concentration rises sharply to 1.37×10^{12} atoms/cm^3 (exceeding the limit). By interpolation, the maximum allowable anneal time before exceeding the 10^{12} atoms/cm^3 threshold is approximately 6.14 seconds.

Table 2: Simulated Al Concentration vs. Post-Implant Anneal Time in Gate-First Flow

Sr. No.	Anneal Time (s)	Al Concentration (atoms/cm^3)	Status / Remark
1	1.0	1.84×10^{-6}	Safe (Negligible)
2	2.0	7.52	Safe
3	3.0	49.41	Safe
4	4.0	148.54	Safe
5	5.0	323.08	Safe
6	6.0	587.84	Safe
7	6.10	1.53×10^6	Safe (Rapid onset)
8	6.12	1.54×10^6	Safe (Pre-threshold)
9	6.14 (Critical)	1.00×10^{12}	Critical Threshold Limit
10	6.15	1.37×10^{12}	Failed (> 10^{12} limit)
11	6.20	1.69×10^{13}	Failed
12	7.0	2.58×10^{14}	Failed
13	8.0	5.22×10^{14}	Failed
14	10.0	9.37×10^{14}	Failed
15	12.0 (Default)	1.24×10^{15}	Severe Contamination

3. QUESTION 1 (C): GATE OXIDE THICKNESS VS. METAL DIFFUSION

(c) Minimum Gate Oxide (SiO2) Thickness Determination:

With the anneal time restored to default, another method to suppress metal diffusion is increasing the gate oxide thickness. The SiO_2 layer acts as a physical diffusion barrier between the aluminum metal gate and the silicon substrate.

The gate oxide thickness was varied from 0.50 nm to 1.00 nm. Table 3 details the simulated Al concentration, and Figure 4 shows the resulting semi-logarithmic curve.

Key Finding:

For oxide thicknesses below 0.81 nm, severe metal diffusion occurs ($> 10^{14}$ atoms/cm^3). Between 0.81 nm and 0.82 nm, the diffusion barrier becomes effective, reducing contamination from 1.42×10^{14} down to 3.89×10^3 atoms/cm^3. Therefore, the minimum gate oxide thickness needed to keep metal diffusion below the 10^{12} atoms/cm^3 limit is approximately 0.818 nm (or ~0.82 nm).

Table 3: Simulated Al Concentration vs. Gate Oxide (SiO2) Thickness in Gate-First Flow

Sr. No.	SiO_2 Thickness (nm)	Al Concentration (atoms/cm^3)	Status / Remark
1	0.50	1.24×10^{15}	Failed (Severe diffusion)
2	0.60	8.27×10^{14}	Failed
3	0.70	5.40×10^{14}	Failed
4	0.80	1.77×10^{14}	Failed
5	0.81	1.42×10^{14}	Failed (Pre-barrier)
6	0.818 (Min.)	1.00×10^{12}	Threshold Limit
7	0.82	3.89×10^3	Safe (Effective barrier)
8	0.85	3.50×10^3	Safe
9	0.90	2.68×10^3	Safe
10	1.00	10.71	Safe (Negligible diffusion)

4. QUESTION 2: GATE LAST PROCESS ANALYSIS (ASSIGNMENT4_GATELAST.ZAM)

The Assignment4_GateLast.zam project was built and analyzed in Semulator3D. In this process flow, the gate stack is deposited AFTER the high-temperature anneal steps for the source and drain.

(a) Al Metal Diffusion After Dummy Anneal:

The sacrificial polysilicon dummy gate is refractory and does not diffuse during source/drain activation. In this project, a dummy anneal was added at the end to represent the cumulative thermal exposure of subsequent process steps. Under the standard dummy anneal at 1000°C, the simulated Al concentration in the active channel is only 1.64 atoms/cm^3, which is virtually zero compared to the 10^{12} atoms/cm^3 threshold.

Table 4 summarizes the complete Process Editor sequence for the Gate-Last (RMG) integration scheme.

Table 4: Process Editor Recipe Steps for Gate-Last (RMG) Flow (Assignment4_GateLast.zam)

Step	Process Stage	Step Name	Material / Chemistry	Operation & Purpose
-10	Substrate & STI	Wafer Setup & STI Isolation	Si / SiO_2 / Si_3N_4	Forms shallow trench isolation and defines active silicon fins/areas
-0.2	Dummy Gate Stack	SacOx, High-k, Dummy Poly	SacOx / HfO_2 / Polysilicon	Deposits sacrificial dummy polysilicon gate over thin sacrificial oxide
-0.4	Dummy Patterning	Lithography & Dummy Gate Etch	Photoresist / Anisotropic RIE	Subtractively patterns dummy polysilicon gate stripes across channel
0.8	Spacer Formation	Conformal Spacer Dep & Etch	Si_3N_4 Nitride Spacer	Forms sidewall dielectric spacers to align subsequent implants
-0.7	S/D Implantation	NMOS & PMOS Source/Drain Doping	As / P (NMOS) & B / BF_2 (PMOS)	Performs heavy source/drain doping shielded by dummy gate
13.9	Activation Anneal	High-Temp Dopant Activation	RTA (> 1000°C Anneal)	Activates S/D dopants with dummy poly in place; no metal exposed
-0.1	ILD & Planarize	ILD Deposition & Oxide CMP	CVD SiO_2 Interlayer Dielectric$_2$	Fills ILD oxide and polishes down stopping exactly on dummy poly
15.3	Poly Pull Etch	Selective Dummy Gate Strip	Selective Wet/Dry RIE Etch	Selectively removes dummy polysilicon, leaving pristine narrow trench
-0.4	Work-Function ALD	PMOS / NMOS Work Function Metal	ALD TiN (PMOS) & TiAl (NMOS)	Conformally deposits work-function metals into narrow recessed trenches
-0.2	Gate Fill & CMP	Bulk Metal Deposition & Metal CMP	Tungsten (W) / Aluminum / CMP	Fills trench, planarizes metal overburden, and isolates gate electrodes
17	Dummy Anneal	BEOL Low-Temp Diffusion Check	Thermal Cycle (1000°C - 1200°C)	Verifies replacement metal gate withstands subsequent BEOL thermal cycles

4. QUESTION 2 (B), (C) & (D): THERMAL MARGIN, COMPLEXITY & HAR CHALLENGES

(b) Maximum Dummy Anneal Temperature:

To test the thermal margin of the replacement gate, the dummy anneal temperature was increased to 1200°C. Even at 1200°C, the Al concentration reached only 3.295×10^4 atoms/cm^3, remaining approximately seven orders of magnitude below the failure limit.

• Maximum temperature before contamination exceeds 10^{12} cm^{-3}: [_____ °C]

(Simulation confirms substantial thermal margin, remaining safe well above 1200°C).

(c) Added Steps and Materials in Gate-Last (RMG) Flow:

The Gate-Last process solves the thermal budget and diffusion problems, but at the expense of added process complexity compared to Gate-First. Extra steps include sacrificial dummy gate deposition, ILD CMP, selective dummy poly pull etch, conformal ALD work-function metal deposition, bulk metal fill, and metal CMP.

Extra materials include sacrificial polysilicon, CMP stop liners (SiN/SiON), and dual work-function metal stacks (TiN, TaN, TiAl, TiAlC).

(d) High-Aspect-Ratio Metal Deposition and Scaling Limit:

A critical requirement of the metal-last process is high-aspect-ratio (HAR) metal deposition. As channel length decreases while gate height is held constant, the aspect ratio (Height / Length) increases significantly. At ultra-narrow gate lengths, precursor pinch-off at the trench opening leads to void formation and incomplete fill.

• Channel length at which process failure / pinch-off occurs: [_____ nm]

Table 5: Comparison between Gate-First and Gate-Last (Replacement Metal Gate) Flows

Category	Gate-First Flow	Gate-Last (Replacement Gate) Flow	Remarks / Impact
Gate Stack Deposition	Deposited before S/D high-temp anneal	Sacrificial poly gate first; metal gate deposited after S/D anneal	Eliminates metal thermal degradation
Thermal Budget Constraint	Severe (Metal must withstand > 1000°C)	High (Metal deposited after high-temp steps)	Allows lower-work-function & low-k materials
Extra Process Steps	None (Conventional self-aligned gate)	Sacrificial poly deposition, ILD CMP, Poly pull etch, Metal fill, Metal CMP	Higher lithography & CMP complexity
Additional Materials	Standard gate metals (TiN, TiAl, W)	Sacrificial Polysilicon, Sacrificial Oxide, ILD Oxide, Polish stop layers	Requires highly selective etch chemistries
High-Aspect-Ratio Challenge	None (Gate patterned subtractively)	Critical (Must void-free fill narrow trenches)	Requires atomic layer deposition (ALD)
Threshold Voltage Stability	V_{th} shift due to metal diffusion & reaction	High V_{th} stability and precise work function tuning	Standard choice for ≤ 14 nm logic nodes

6. OBSERVATIONS & CONCLUSION

Observations:

1. In Gate-First CMOS, the default 12 s activation anneal caused severe aluminum diffusion into the active silicon channel (1.24×10^{15} atoms/cm^3), exceeding the 10^{12} atoms/cm^3 operational limit.

2. Reducing the anneal duration suppressed metal contamination, establishing a maximum safe anneal time of approximately 6.14 seconds.

3. Increasing gate oxide thickness demonstrated that SiO_2 acts as an effective diffusion barrier; thicknesses ≥ 0.818 nm successfully suppressed contamination below 3.89×10^3 atoms/cm^3.

4. In Gate-Last CMOS, the refractory polysilicon dummy gate completely shields the active region during thermal anneals. Simulated contamination remained negligible (1.64 atoms/cm^3 at 1000°C and 3.3×10^4 atoms/cm^3 at 1200°C), proving superior thermal stability.

5. Gate-Last flows achieve high thermal resilience at the expense of higher complexity: requiring sacrificial gate patterning, ILD CMP, selective poly pull, and high-aspect-ratio metal deposition.

6. Analysis of the 14 nm FinFET process confirmed the universal industry adoption of Gate-Last (RMG) architectures, utilizing TiAl for NMOS and TiN for PMOS to achieve symmetric threshold voltages.

Conclusion:

The trade-offs between Gate-First and Gate-Last integration schemes were evaluated using Semulator3D. While Gate-First suffers from severe thermal budget limitations due to metal cross-diffusion during source/drain activation anneals, the Gate-Last (RMG) flow decouples high-temperature anneals from gate stack deposition, enabling low-leakage, high-performance nano-scale CMOS devices.